

# CORNERSTONE PDK Monitor — User Manual v1.0.3

## CORNERSTONE PDK Monitor — User Manual

**Version:** v1.0.3 **Single-file browser dashboard for silicon photonics PDK management.**  
Designed by Emre Kaplan · [pdk.cornerstone.soton.ac.uk](mailto:pdk.cornerstone.soton.ac.uk) Licence: TAPR Open Hardware  
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### 1. What this is

CORNERSTONE PDK Monitor is a one-page dashboard that keeps every artefact in a silicon photonics process design kit visible, browsable, and maintainable from a single browser tab. It runs entirely offline as one self-contained HTML file: no server, no install, no toolchain.

A PDK is a moving target. Layout files, S-parameter sweeps, fab process documents, EDA wrappers, post-processing scripts, supplier test reports, and design notes accumulate across years and across teams. The Monitor pulls all of that into one consistent shape: each **Technology Platform** holds a list of **Building Blocks (BBs)** — grating couplers, MMI splitters, ring filters, modulators, etc. — and every BB carries its own layout, simulation data, scripts, tests, fab notes, EDA links, variants, and documents.

A second tab kind, **DOCS**, gives the same BB structure to design manuals, app notes, fab guides, and supplier documents, so a design manual chapter behaves like a first-class entity rather than a loose file attachment.

Everything you build inside the dashboard is held in a single JSON document (dashboard.json). You can export it, share it, version it, or pull a colleague's view in and merge it with your own.

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### 2. Quick start

1. **Open** `cornerstone-pdk-monitor-v1.0.3.html` in Chrome, Firefox, Safari, or Edge. No server required.
2. **Accept** the licence terms on first launch.
3. The dashboard loads pre-populated with the full CORNERSTONE BB set. Browse tabs, click any BB card to open its detail view.
4. To **sign in**, click the user icon in the header. Three options:
  - **Read-only** — browse and search; no edits.
  - **User-admin / Admin password** — typed identity + guest or admin password.

- **GitHub PAT** — paste a Personal Access Token; the dashboard auto-fills your dashboard identity from your GitHub login and promotes you to user-admin. No password handout needed.
5. To **start from a clean slate**, click *Import* in the header and pick the bundled dashboard.json, then choose *Replace whole dashboard*.
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### 3. The dashboard layout

#### Header

- **Tabs strip** — one tab per Technology Platform or DOCS bundle. Each shows a TECH or DOCS pill. The strip stays a single fixed-height row no matter how many tabs you add — once the tabs overflow the viewport, a thin horizontal scrollbar appears (the + **Add tab** button never wraps to a second row).
- **Global search box** with cross-tab dropdown.
- **Tool buttons** — Import, Export, Manage Lists, Distribution, the Activity Log, the Insight agent panel, and the user/sign-in menu.

#### Body

The active tab. Each tab holds groups (sub-entities, sometimes called platforms); each group renders as a card with rows of BB cards inside.

#### Per-platform-group head buttons

Every group head carries an action row:

- **↺ Sync** — replays every previously-used GitHub URL on every BB.
- **Design manuals** — opens the platform-level design-manuals window (see section 5).
- **⬇ Import platform** — opens the unified import modal (sources + sections + token, all in one place — see section 4).
- **+ Add sub-entity** — creates a new group.

#### Platform-level Design manuals

Every Technology Platform group head carries a **Design manuals** chip with a live file count. Click it to open the design-manuals window — a focused view of every platform-level document attached to that group:

- **↺ Sync from GitHub** — pulls every PDF / DOC / DOCX / MD / TXT from the platform's docs/, doc/, documents/, documentation/, manuals/, datasheets/ folders on the configured repo (recursively), plus README / datasheet / overview files at the platform root.
- **📁 Import (upload)** — opens a file picker for local uploads.
- **⚙ Configure source** — set a GitHub URL specifically for these manuals (separate from the platform's main repo URL — useful when manuals live in a different repo).
- Each row carries download, PDF preview, pin, and delete.

- Files dedupe by filename — re-sync replaces in place, preserving id, pin state, and uploader.

## Footer

Version stamp, status counts, storage indicator.

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## 4. Importing from GitHub (unified modal)

Click **↓ Import platform** on any group head. The modal is one place for everything:

### Sources

Manage as many GitHub URLs per group as you want. Each row has an on/off tickbox, an editable label, the URL, the components subdirectory, a ↗ test button, and × remove. Add via **+ Add** (blank row) or **+ Paste URL** (auto-derives the label). **All / None** to flip the ticks at once.

| Enabled sources  | Importer used          | BB naming                                                                                                                                       |
|------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b>         | Platform-walk importer | BB names come from *.gds filenames inside components/. Source label is a friendly tag — it never touches BB names.                              |
| <b>2 or more</b> | Name-match importer    | Each source's label is the target BB name. Files at that URL route into the BB whose name matches the label (case- and underscore-insensitive). |

### Sections to fetch

Six section ticks (Layout / S-matrix / Tests / Fabrication / Scripts / Documents). All / None / Light presets. **Edit keywords** opens a modal where you tune the folder-name keywords used to recognise each section (e.g. add meas to the Tests keyword list). Edits persist in MODEL.importFolderAliases and travel inside dashboard.json.

### Token + diagnostics

The PAT field has a **Test token + sources** button that runs a two-step diagnostic:

- Calls /user to verify the token is valid and reports who it belongs to + scopes / rate-limit.
- For every enabled source, calls /repos/{owner}/{repo} and reports ✓ read · ✓ push · — admin (or the failures) per repo. Surfaces SSO-authorisation links directly from the X-GitHub-SSO response header when the org enforces SSO.

A cheat-sheet below the rows explains exactly which token-page checkbox to tick when you see  $\times$  push (classic PAT scope, fine-grained PAT permission, org-level fine-grained PAT enablement, SSO).

### Footer

- **Cancel** — close without saving.
- **Save sources** — persist source-row edits without running an import.
- **Import enabled** — runs the import (against every ticked source in turn). Per-source progress streams into a log panel under the modal.

The **Test token + sources** button mirrors the same diagnostic inside the Distribution panel.

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## 5. Working with Building Blocks

Click any BB row to open its detail panel. The tab list is:

Details · Layout · Docs · Scripts · EDA · Fabrication · S-matrix · Tests · Variants · AI Insight

### Layout

- Auto-Sketch (generated from BB parameters) or uploaded .gds / .oas / .svg.
- Upload, fetch from GitHub, open PR-publish.
- Measurement and scale calibration tools.
- **YAML descriptor card** below the layout viewer — surfaces the canonical BB descriptor (the file that ships with the GDS in components/) with inline preview + download.

### Docs

- All design manuals, datasheets, app notes, READMEs for the BB.
- **+ Upload document** (PDF / DOC / DOCX / MD / TXT, 50 MB cap each).
- **↓ GitHub fetch** — paste a repo docs/ URL; every PDF / DOC / DOCX / MD / TXT inside is pulled into the BB's documents. YAML files are skipped (they belong on the Layout tab). Dedupes by filename.

### Scripts

Pyodide-backed runner. Each script has its own card; click ► to run. Globals tests (parsed test data) and data (S-matrix datasets) are exposed automatically. To emit a chart, write your figure to a buffer and print("DASH\_PLOT\_PNG:" + b64(...)) — the emit(fig) helper does it for you. Output streams below the editor, with PNG previews inline.

### Plots

When a script produces a plot:

- **No figure windows open yet** → each plot pops automatically into its own new browser window.
- **Some figure windows already open** → an inline prompt appears at the top of the output modal: *“3 new plots ready. Choose where to send them all: [ ↗ New figure windows (3) ] [ Merge into Plot 1 (2 plots) ] [ Skip ]”*
- Per plot, three buttons:
  - ↗ **New figure window** — fresh popout, fresh id.
  - **Merge into...** — opens an inline picker of every open figure window; click to append the plot into that window.
  - ⬇ **Download PNG**.

Figure windows stay open until you close them manually. Compact layout (small header, hover-only action buttons in desktop mode, always-on in narrow window mode) so a minimised window still shows the plot clearly.

### Data (S-matrix / Tests)

S-parameter datasets, sweep files (TXT, S2P, CSV, NPZ), and parsed metadata. Click **Plot raw** on any dataset to open the chart; the chart toolbar has:

- Y-axis Linear / dB toggle (preserves overlays).
- **+ Overlay dataset...** — opens a floating draggable picker listing every parsed dataset on every BB across every tab. Click to add as extra traces in-place — chart updates instantly. Repeat as needed.
- **Clear overlays** — drops everything except the first dataset.
- Inline legend showing how many datasets / traces are visible.

When you click **Plot raw** on a different BB while a chart is already open, a small overlay asks: *“ Merge into open plot” / “ ↗ New plot window” / “Cancel”*.

### Subfolders inside folders

Test folders and S-matrix folders support unlimited-depth nesting. Each folder has a **+ Subfolder** button that creates a child folder; child folders render indented with a left border. The full folder UI (Add files, Plot all, Run script, Rename, ×, pin) works at any depth. The × delete walks the entire tree, so subfolders delete correctly from wherever they sit in the nest.

### Fab

Sub-tabs per fab process step or supplier.

### EDA

Links to external EDA tool definitions. Pin to keep the wrappers you actually use at the top.

## AI Insight

A built-in chat panel that can read the files attached to the current BB and answer questions. Configure your Anthropic / OpenAI API key once via the ⚙️ Configure button.

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## 6. Distribution panel

The team-sync hub. Five sections.

### 1. Sources

List of GitHub repos to fetch from. Each row has a tickbox, label, repo, branch, path (dashboard.json for the main, users/<id>/dashboard.json for slices).

The **Test token + sources** button next to the PAT runs the same two-step diagnostic as the import modal — confirms the token is valid and reports per-source read / push / admin permissions.

### 2. Fetch & adopt

- **Fetch all enabled sources** — pulls every ticked source.
- **Discover user slices** — walks the configured users/ folder and lets you pick which users/<id>/dashboard.json slices to fetch.
- **Discover forks** — walks every fork of every enabled upstream source on GitHub, probes each fork for a dashboard.json (preferring users/<fork-owner>/dashboard.json, falling back to root), then shows a floating picker where you tick which forks to add as Distribution sources. Useful for pulling contributions from contributors who work in forks without filing a PR.
- **Fetch my slice** — fetches just your own users/<id>/dashboard.json. Bootstraps the folder if it doesn't exist yet.
- **Merge into my dashboard / Replace whole dashboard** — apply the fetched payload.

### 3. Publish

- **Publish as Pull Request** — opens the rich **PR-config overlay**: editable title, body (markdown textarea), base / head branches, reviewers, labels, draft toggle. The Distribution panel + its log stay visible behind the overlay. Submit → posts POST /pulls → shows the PR URL in the log → attaches labels (POST /issues/{n}/labels) and reviewers (POST /pulls/{n}/requested\_reviewers).
- **< Direct commit** — preflight checks push permission first. If your PAT can't push to the upstream, you get a modal: “⚠️ No push permission. [ Fork & publish instead]”. One click forks, commits to your fork, and prints a compare URL + clickable **Open PR to upstream** button (uses the same PR-config overlay).
- **Fork & publish** — explicit version of the same flow.
- **Open PR to upstream** (appears after Fork & publish) — opens the rich PR-config overlay; on submit calls POST /pulls. Falls back to a one-click **Open PR creation**

**page on github.com** button when the API call fails (e.g. fine-grained PAT lacks Pull requests permission on the upstream).

Rate limit: flat **30 s** between successful submits, **20 / hour** per identity (both for PRs and direct commits). Admin role is exempt.

#### 4. GitHub collaborators (live)

Click **Show collaborators** to load three live sections:

- **Collaborators** — per row: avatar, login, permission chip, PR count, and a × remove button (admins only).
- **Open Pull Requests** — every open PR on the repo (up to 50), with author, fork source (if any), last-update date, draft flag, a ↗ Review link, and a Merge... button (admins only) that opens the PR's merge dialog on github.com.
- **Invite collaborator** (admins only) — type a GitHub username, pick a permission level, click **Send invite**. Sends a real GitHub invitation.

#### 5. Resolve duplicates

The existing duplicate-resolver tool — collapse BBs / platforms / tabs that ended up with the same name from different contributors.

#### Sign in with GitHub PAT (sign-in modal)

A new section in the sign-in modal lets you authenticate with just a PAT — no dashboard password. The dashboard:

1. Validates the token, reads your GitHub login.
2. Promotes you to user-admin with your GitHub login as your identity.
3. **Auto-discovers your fork(s)** of every configured Distribution source and adds them as new (enabled) source rows, preferring the path `users/<your-login>/dashboard.json` so the default fetch pulls your slice from your fork.

If a fork-row with the same (repo, branch, path) tuple already exists, it isn't re-added — sign-in is idempotent.

A **How do I get a Personal Access Token?** accordion under the PAT field gives step-by-step PAT creation + scope guidance for both classic and fine-grained PATs, plus an SSO-org note.

If you forget the admin password, the sign-in modal has a **🔑 Reset admin password to factory default (this browser)** link that wipes the local security record. Other browsers / GitHub repos are untouched.

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## 7. Search and keyboard

- / or Cmd/Ctrl + K — focus the global search box.
  - t — switch to the next tab (wraps).
  - n — add a new BB to the first group of the active tab.
  - ? — open the keyboard-shortcut cheatsheet.
  - Esc — close the active modal, dropdown, or floating window.
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## 8. Mobile / tablet

Two responsive breakpoints:

- **≤ 900 px** — header wraps, search drops to its own row, group cards stack one per column.
  - **≤ 600 px** — tabs scroll horizontally, modal goes near-fullscreen, H1 + stats hide to free up real estate.
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## 9. Importing, exporting, distribution

### Export

Click *Export* in the header — full state as dashboard.json.

### Import

Click *Import* and choose a JSON file. Merge (additive, with diff preview) or *Replace whole dashboard* (auto-backup written first). Every import path schema-validates the incoming payload.

### Distribution

See section 6 above.

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## 10. Undo, Activity Log, and auditability

30-second Undo on every destructive action. Activity Log button in the header shows the last 100 mutations (who / when / what / item). Auto-backup before any *Replace whole dashboard*.

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## 11. Statistics

The header chart and the *Statistics* expander aggregate counts across every tab marked **TECH**. DOCS-tabbed bundles are excluded from BB stats.



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## 12. Troubleshooting

**Sign-in fails in normal Chrome but works in incognito.** The admin password was probably changed locally; use the  **Reset to factory default** link in the sign-in modal.

**“PR open failed — 403 — Resource not accessible by personal access token”** — fine-grained PAT missing **Pull requests: Read and write** on the upstream repo, OR the upstream org has fine-grained PATs disabled. The fork + commit already succeeded — click the **Open PR creation page on github.com** button surfaced under the error to finish via github.com (one click).

**Push permission preflight fails on direct commit** → the dashboard offers *Fork & publish instead*. One click moves the contribution to your fork + opens a PR back to upstream.

**Plot windows close after running a new script** → fixed in v1.0.3. Each popout uses a unique window name embedding a timestamp so the browser never reuses a window. They stay open until you close them.

**Subfolder × button does nothing** → fixed in v1.0.3. Delete walks the entire folder tree.

**Folder keyword doesn’t match a custom folder name in your repo** → click **Edit keywords** inside the import modal and add your folder name to the matching section.

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## 13. Where things live on disk

- `cornerstone-pdk-monitor-v1.0.3.html` — the only file you need to open. Contains the seed dashboard, all app code, every dependency.
  - `dashboard.json` — full state restore-point.
  - `USER_MANUAL.md` / `.docx` / `.pdf` — this document.
  - `CHANGELOG.md` — release notes.
  - `CORNERSTONE_PDK_Monitor_Introduction.pptx` — slide deck.
  - `README.txt` — short release readme.
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## 14. AI Insight quick start

1. Click **Insight** in the header.
  2. Click  **Configure** and paste your Anthropic or OpenAI API key.
  3. Click **Chat with the agent** to open a thread on the current BB or DOCS entry.
  4. Expand ► **Data files in context** to attach any of the BB’s CSV / JSON / S-matrix files. The agent can run quick analyses on them via *Analyse files with AI*.
  5. Click **Detach** to pop the chat into a floating window.
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## 15. Licensing and credits

CORNERSTONE PDK Monitor is licenced under the **TAPR Open Hardware Licence v1.0**.

Designed and maintained by **Emre Kaplan** at the **CORNERSTONE silicon photonics platform**, University of Southampton.

- Project home: <https://pdk.cornerstone.soton.ac.uk>
- Open platform repo: <https://github.com/cornerstone-uos/cornerstone-community>
- Main PDK repo: <https://github.com/cornerstone-uos/cornerstone-pdk>
- Website: <https://cornerstone.sotonfab.co.uk>

Bug reports, feature requests, and pull requests welcome at the GitHub repos above.